

# Mastering Multi-Cloud AI Deployments with Open Source Tools

As AI adoption accelerates, enterprises are increasingly embracing multi-cloud strategies to leverage the unique strengths of different cloud providers. But managing these complex environments can be daunting. Open source tools offer a compelling solution, providing cost-effective, customisable, and vendor-neutral options for deployment, orchestration, monitoring, and governance. Let's explore the advantages of multi-cloud for AI, the key open source tools, and best practices for success.



**M**ulti-cloud architecture is the practice of distributing workloads and applications across multiple cloud providers. This approach allows organisations to leverage the strengths and capabilities of different cloud platforms, mitigating the risks associated with vendor lock-in, and increasing flexibility and scalability. While it brings vendor diversity by utilising multiple cloud providers, organisations are exposed to the risk of service disruptions when

managing complex environments. Most organisations still opt for it, as choosing the right product and platform improves performance in a geographically distributed cloud infrastructure of each provider. Also, it can lead to cost optimisation by choosing the right workload management with proper allocation of resources.

## Navigating the multi-cloud landscape for AI

In the AI era, businesses are turning to

multiple cloud providers to gain:

- **Flexibility:** Access a wider range of services and features tailored to specific AI tasks
- **Scalability:** Efficiently handle fluctuating workloads, from massive training jobs to real-time inferences
- **Cost optimisation:** Choose the most cost-effective solutions for different needs
- **Availability:** Minimise downtime and maximise data security by distributing it across regions and providers